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Smartphone Application As An Air Quality Monitor Using Raspberry Pi For Reducing Air Pollution

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***Abstract*— Good air quality is very important in human health. However, air pollution has been polluted everywhere, whether in the house, in cities. This is caused by a growing industry that produces air pollution. Room air pollution and various cities will contribute to air pollution, which will cause problems for human health. This work measures the carbon monoxide level using the MQ-7 sensor and measures the air's carbon dioxide level with the CCS811 sensor and MH-Z19 sensor. All three sensors are connected to the NodeMCU ESP8266, and we call it a NodeMCU module. Tests are carried out in three rooms in the house, and then three NodeMCU modules are used so that the three modules will send data alternately. Raspberry Pi3 (RPi3) functions as a server that collects data and acquires information data from sensors. Data from the CCS811 (TCOV) sensor, MH-Z19 (co2), will be processed first with fuzzy logic rules to make decisions about CO2, which are right, moderate, nasty conditions. The process is carried out on the Raspberry Pi 3 server with MQTT using publication, customer, fuzzy logic, and broker services. The value of the sensor's measurement results can be accessed via a smartphone and provides good air control values.**

Keywords; Indoor Air Quality; Raspberry Pi 3; MQ7, MH-Z19, CCS811sensor; module NodeMCU; NodeMCU ESP8266;

I. Introduction

Air pollution is a mixture of colorless and odorless gases (such as oxygen and nitrogen) that fill the space above the earth as we breathe when we breathe [1]. A healthy environment is very influential on the physical health of living things. An essential factor supporting a healthy environment is air quality that meets health standards. Air contains oxygen needed for life. But besides oxygen, there are other substances in the air, such as carbon monoxide, carbon dioxide, formaldehyde. Certain limits on these substances' levels can still be neutralized, but it can interfere with health if it exceeds normal limits. Good or bad air quality is influenced by daily human activities [2] [3].

The polluted air polluted in the environment and in the room can also be contaminated by harmful chemicals and other materials [4]. Air pollution can cause various diseases such as asthma, wet lungs, and even coronary heart disease. Pollution can be done outdoors or indoors. However, people spend around 89% of their activities indoors, such as at work, home, school, etc. [5]. CO2 is one of the highest indoor environment elements due to respiration and human activity in the room. High CO2 levels create various irritants and reduce cognitive performance [6]. Monitoring air quality in the room needs to be done to determine the air quality in the room. Indoor air quality monitoring can ensure that the indoor environment is safe for staying or doing activities [6]. With current technology, indoor air quality monitoring can also be integrated with the Internet. The concept of the Internet can be implemented on indoor air quality monitoring systems. Internet of Things is a technology that can make things smarter than before [7][8].

II. Related Works

Indoor air quality monitoring systems using CO2 sensors, CO sensors, and raspberry pi 3 with the system output are DC fans, but the DC fan feature is only in ON / OFF state [9]. Users can manually turn on the ventilation fan from the smartphone application when the air quality tolerance limit is exceeded. Their suggestion for further research is that the system can be improved with various sensors and increase its reliability and accuracy. The system also has no automation to turn on the ventilation fan when the gas exceeds the limit. Others learn about indoor air quality monitoring; Kumar created an indoor quality monitoring system that is integrated with servers in the cloud. Sensor data is displayed in graphical form on the cloud server dashboard. The MQTT protocol is essential for building communication between the raspberry pi and the cloud server. The medium used to connect from the indoor air quality monitoring system to the Internet uses a WiFi access point [10]. The Active Kumar System uses various parameters for monitoring. There are PM 2.5, carbon monoxide, carbon dioxide, temperature, humidity, and air pressure. But the system made by Kumar does not have an actuator to release bad air in the sensor environment. Vivek proposed a ducting system using fuzzy logic control to distribute the air in the room. There are two inputs of temperature and humidity from the analysis of fuzzy logic. The fuzzy output is to control the fan's speed, pump, and channel in the fan. The system is using fuzzy Mamdani. This system functions appropriately to control the temperature and humidity of the room. But the Vivek System does not have a monitoring dashboard for real-time environmental monitoring. This system uses Matlab to compare fuzzy code with simulation [11].

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III. RESEARCH METHODOLOGY

Indoor monitoring is designed using sensors that can measure carbon monoxide levels by using MQ-7 sensors and measuring carbon dioxide levels in the air with CCS811 sensors and MH-Z19 sensors. All three sensors are connected to NodeMCU ESP8266 [11] (NodeMCU module). The proposed system architecture has 3 main parts of the system, Fig1. Namely, the NodeMCU module as an indoor sensor, the Raspberry Pi3 (Rpi3), as a server, acquires information data from the sensors in the MQTT broker, the Smartphone Dashboard. The three NodeMCU modules are connected to the MQTT broker on the RPi3 server. The sensors in each NodeMCU module will measure analog data, and the data is sent to customers through MQTT brokers [12]. The fuzzy logic process to determine CO2 decisions (Good, Medium, Less) is done with two inputs from the CCS811 sensor and the MH-Z19 sensor. The Fuzzy, Publisher, Broker, Subscriber process is carried out, and the MQTT broker determines the output.

IV. DESIGN AND ANALYSIS

RPi3 will receive information from carbon monoxide sensors (MQ-7) to measure carbon monoxide levels in the air and CCS811 (TCOV). MH-Z19 (co2) is used to measure carbon dioxide (CO2) data in the form of analog signal data, and analog sensor data will be converted into digital data at Rpi3. The proposed indoor air quality monitoring system has a fuzzy logic control for output (good, medium, labor) automatically for CO2. Fuzzy logic input data is from digital data that is sent from the CSS811 sensor and the MH-Z19 sensor. The result of defuzzification is the interval value to determine CO2 data's accuracy from the two sensors. Sensor data will be sent/saved to the server and provided as a dashboard for monitoring purposes. Information is sent by the MQTT (Message Queuing Telemetry Transport) protocol. Raspberry pi3 will publish CO2 and CO data strings to MQTT Brokers on the RPI server using the Internet. The dashboard uses Python, Broker server. The dashboard consists of CO, CO2, Real-time CO2 value charts, and CO2 values in the room environment. Users can access dashboards from browsers and smartphone devices. Below is a flowchart of the proposed system.

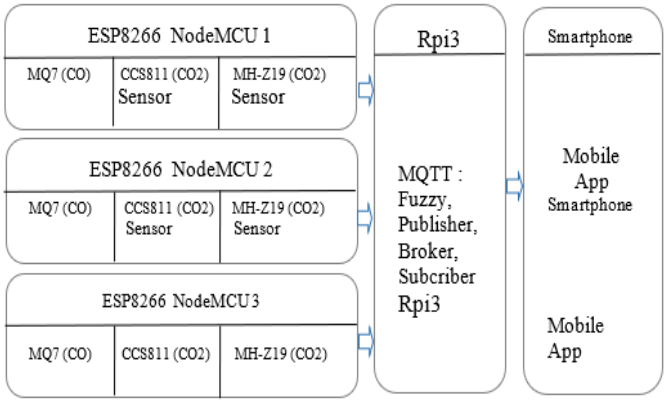

Fig. 1. System Architecture Overview

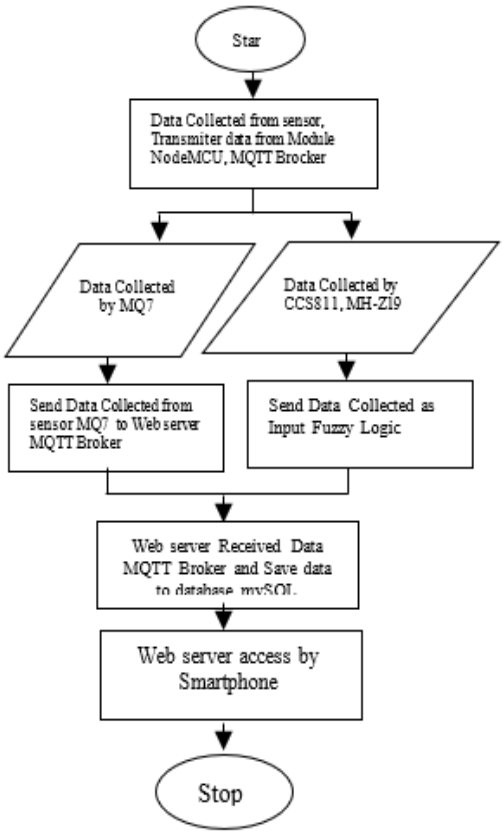

Fig. 2. Flowchart of Proposed System

A. Hardware Design

This system uses a variety of hardware and sensors that are connected. The Esp8266 MCU node module has a port that is connected to all three sensors. Fig 3 explains the connection of the sensor pin. The MQ-7 sensor is connected to port D5 (digital output pin-D5), MH-Z19 sensor with GPIO12 (D6), GPIO13 (D7), and CCS811 (TCOV) sensors connected to GPIO3 (D1), GPIO4 (D2) ports of ESP8266 NodeMCU (Module NodeMCU). The NodeMCU module will send data to MQTT Server at RPi 3.

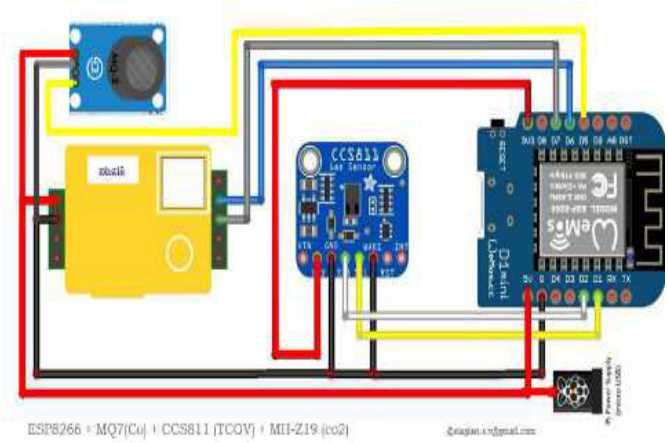

Fig. 3. Module NodeMCU Architecture System

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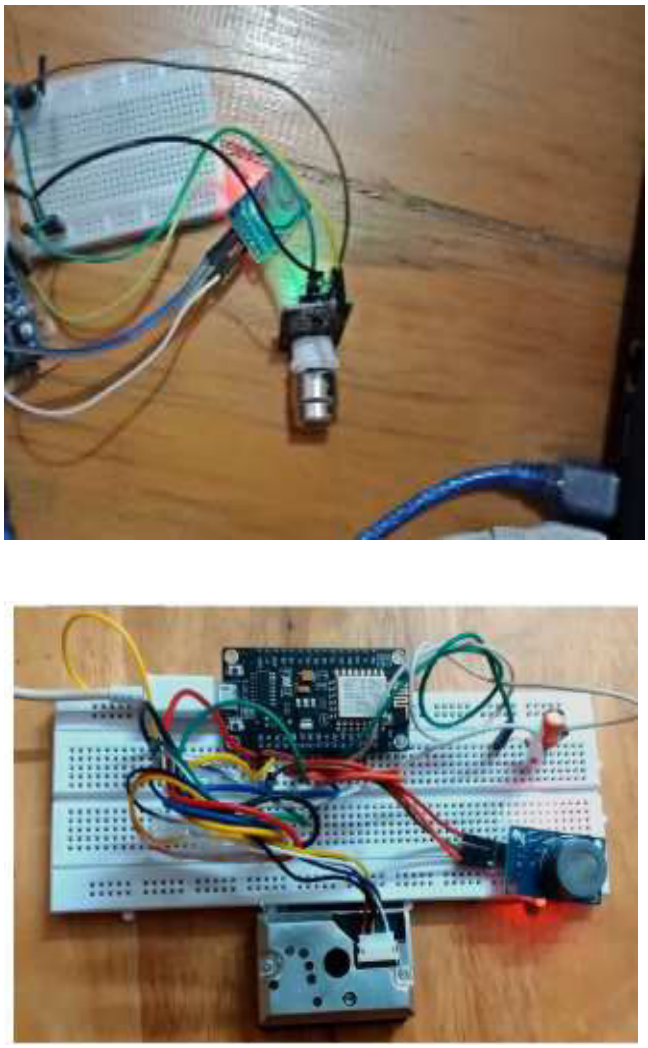

Fig. 4. Implementation Module NodeMCU

B. Rpi3 server

Server Design with Rpi3 aims to hold data displaying sensor data. ESP8266 NodeMCU (Module NodeMCU) and three NodeMCU Modules that function as data senders via MQTT clients to connect to local brokers (Rpi3) with dynamic IPs (ESP8266 clients) through 192.168 host -h. 43.0 or -h raspberry Pi local. Three NodeMCU Modules are connected to the server using the IP host, shown in Table 1.

Raspberry pi 3 receives data streams from the NodeMCUs Module, and local MQTT brokers can be run with local IP or -127.0.0.1 on systems that have built MySQL tables as a data container, and the data can be accessed with the address -h 192.168. 43,246.

Basic communication between ESP8266 NodeMCUs (Module NodeMCUs) with Raspberry pi3 as the MQTT Broker server can subscribe and publish. This system uses Block hardware to publish sensor data using the Internet with the MQTT Protocol and the Secure Socket Layer (SSL) cryptographic protocol. The MQTT broker's server accepts sensor data via the NodeMCUs Module and will be stored in my SQL database, also indexing the data, so the data can be displayed on the dashboard that will be accessed by the user. The dashboard server can display sensor data and describe the state of the room conditions.

Sensor data will be received in real-time, and data stored continuously on the MQTT Server using the MQTT protocol and the Secure Socket Layer (SSL) cryptographic protocol. So that no ambiguous data is stored between the required data and security of the data is maintained.

TABLE I. HOST MQTT CLIENT

Indoor Sensors	Host
Module NodeMCU 1	192.168. 40.241
Module NodeMCU 2	192.168. 40.242
Module NodeMCU 3	192.168. 40.243
MQTT Broker (Server Rpi3)	192.168. 43.246

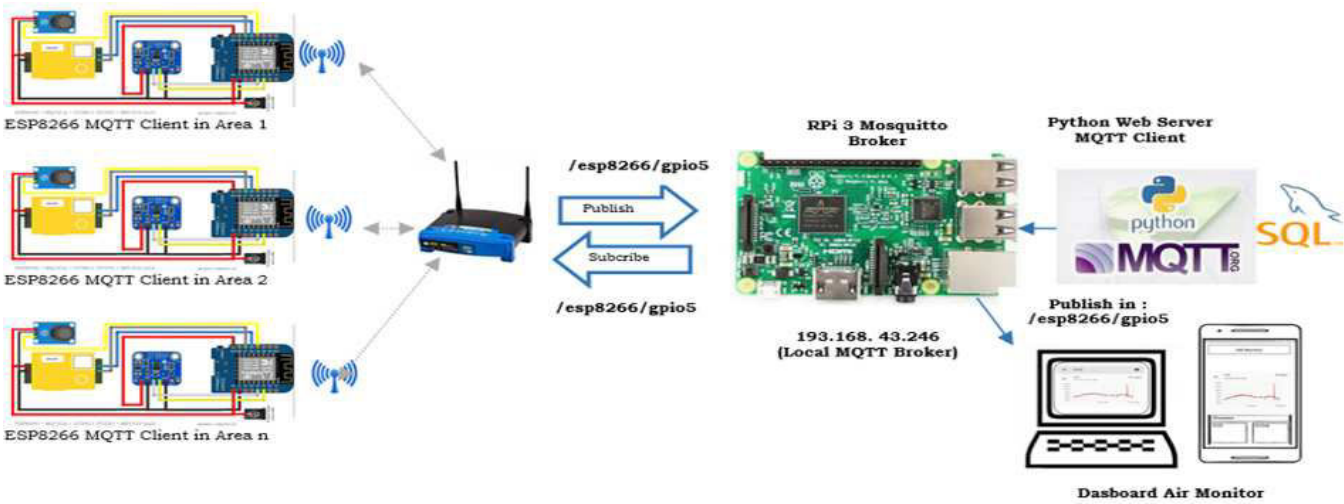

Fig. 5. Design System

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C. *System testing*

This system was built to be able to index the air quality of each room. Figure 6 shows the results of measuring the system using a smartphone. The system will display data every 15 minutes [13]. At the same time, the graph data will work periodically and also display data CO = 3, CO2 = 26, and fuzzy logic displays the results of fuzzification data in the notification "Very Good Air Quality".

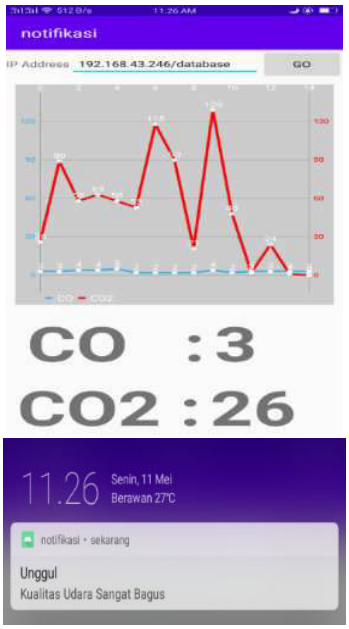
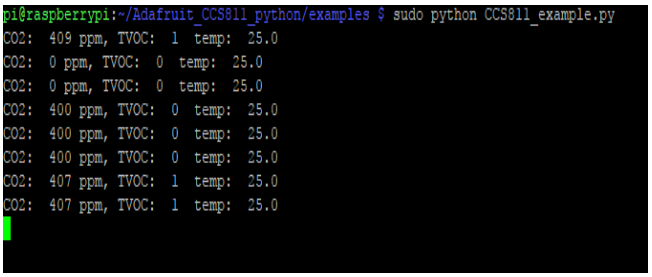
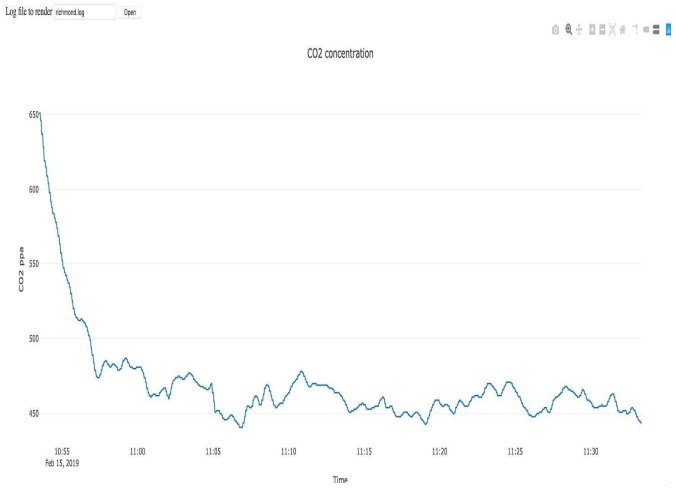


Fig. 6. Smartphone access data sensor

Data on each sensor can also be accessed via the RPI3 server, and graphs can also be accessed via a web server that is built by the MQTT broker. In figure 7 (a), there is data displayed for the CCS811 sensor and also the graph. In addition, with the same command, we can also access data for each sensor.



(a) Acess data Sensor CCS811 (TCOV)



(b) Graf data sensor CO2

Fig. 7. Acess Data sensor

Measurements were made for three different rooms, then the results we can see get different data from each room. The results of each room can be seen in table 2. The data periods shown are 10:00 - 12.00.

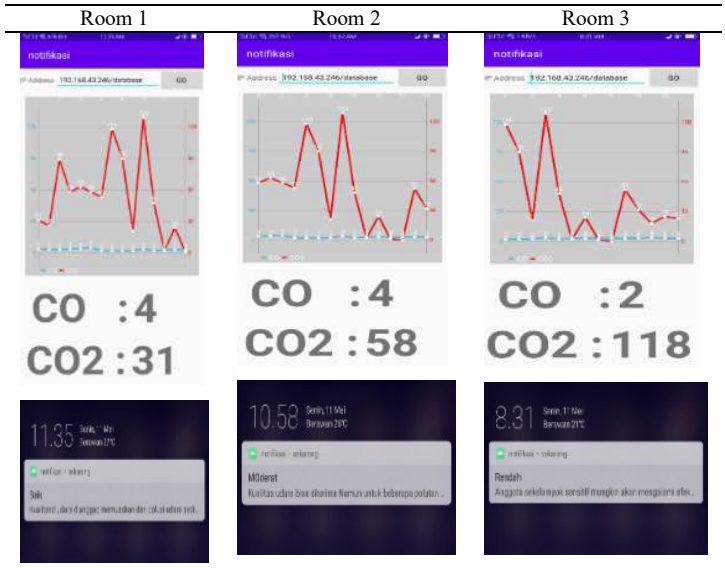


TABLE II. RESULT TESTING

Time	Room 1		Room 2		Room 3	
	CO	CO2	CO	CO2	CO	CO2
10.00	4	26	3	25	2	26
10.15	2	28	3	27	3	26
10.30	3	25	4	27	3	27
11.00	3	26	3	26	2	25
11.15	3	26	4	110	3	150
11.26	3	26	2.5	27	4	28
11.30	3	26	3	27	3	28
12.00	3	26	3	27	3	28

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V. CONCLUSION

Current technology, indoor air quality monitoring can also be integrated with the Internet. The concept of the Internet can be implemented on indoor air quality monitoring systems. This work measures the carbon monoxide level using the MQ-7 sensor and measures the level of carbon dioxide in the air with the CCS811 sensor and MH-Z19 sensor. And all three sensors are connected to the NodeMCU ESP8266, and we call it a NodeMCU module. Tests are carried out in three rooms in the house. Then three NodeMCU modules are used so that the three modules will send data alternately. Users can view notification content and CO, CO2 conditions in each room using a mobile application. Mobile application view results from the detection of air quality anywhere because this design integrates with IoT devices.

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